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GEOLOGY

Paleomagnetism of the Kiselevka Complex of the Kiselevka–Manoma Terrain (Sikhote-Alin): Geodynamic Consequences

A. N. Didenko[a], Academician of the RAS A. I. Khanchuk[b], and A. I. Tikhomirova[a]

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According to the present-day geodynamic concepts [3, 10], the Sikhote-Alin–North Sakhalin orogenic belt is represented by fragments (terrains) of Jurassic–Early Cretaceous accretionary prisms, Early Cretaceous island arcs, and turbidite basins of the active margin of the Eurasian paleocontinent. According to these models, terrains formed in paleozones in different geodynamic settings and then were juxtaposed as a result of large-scale horizontal displacements, forming the modern orogenic belt.

According to another alternative model, describing the evolution of the Sikhote-Alin–North Sakhalin orogenic belt, "the variety of structural and material complexes developed along the continental margin is consequence of continental riftogenesis of the continental crust, accompanied by tectonic dislocations, rather than accretion of the oceanic crust onto the continental margin" [8]. Moreover, "the distinct tectonomagmatic interdependence between ... rift grabens and horst-accretionary systems do not allow us to consider these structures ... as terrains formed independently of each other..." [9].

The reliability of one model or the other can be confirmed by evaluation of the true scale of horizontal displacements of blocks (terrains) comprising the Sikhote-Alin margin of Eurasia using reliable paleomagnetic data. It follows that the task of obtaining the data on the primary paleomagnetic directions for Mesozoic rocks of Sikhote-Alin is very important and relevant.

[a] *Institute of Tectonics and Geophysics, Far East Branch, Russian Academy of Sciences, ul. Kim Yu Chena 65, Khabarovsk, 680063 Russia*
[b] *Far East Geological Institute, Far East Branch, Russian Academy of Sciences, pr. Stoletiya Vladivostoka 159, Vladivostok, 690022 Russia*
e-mail: alexei_didenko@mail.ru

This work presents the first paleomagnetic data obtained for Lower Cretaceous rocks of the Kiselevka complex of the Kiselevka–Manoma terrain (Fig. 1a). For a long time, this terrain was regarded as an Early Cretaceous accretionary complex, comprising fragments of Jurassic basalts and cherts [7]. In later works, the complex is considered as Middle Cretaceous (Aptian-Albian) turbidite matrix; Early Cretaceous (pre-Aptian) rocks are referred to oceanic formations [4, 11]. In addition, the Kiselevka–Manoma accretionary complex comprises fragments of the Hauterivian–Barremian island arc [6].

On the left-hand side of the lower reaches of the Amur River, to the southwest of the village of Kiselevka, the structure of the volcanic-siliceous complex studied (Fig. 1b) is a system comprised of four tectonic plates made of banded and massive cherts, alternating with bodies of basaltic lava.

The strike of beds in tectonic plates is northwest; the dip angles are 20°–80°. Basalts are represented predominantly by types with a SiO_2 content of 47–50%; picrobasalts and andesibasalts are less common. All types of basalts belong to sub-alkaline and normal alkalinity series [2, 3]. At the present time, the age of the basalt association is regarded as Early Valanginian–Middle Barremian (~135 Ma) [5].

For paleomagnetic analysis, we took 183 oriented samples of sedimentary–volcanogenic rocks (basalts, tuffs, aleurolites, and cherts) from 17 sampling sites within the Kiselevka terrain (Fig. 1b) (51.40°N, 138.95°E). Samples were oriented using a mountain compass; the magnetic inclination in the area of study (–12.2°) was taken into consideration in calculation of the mean paleomagnetic directions. As a result of step thermodemagnetization (thermal cleaning) of samples and their processing using standard software for paleomagnetic research, it was established that majority of samples collected have satisfactory paleomagnetic stability. In these samples two important components of statistical significance (low-tempera-

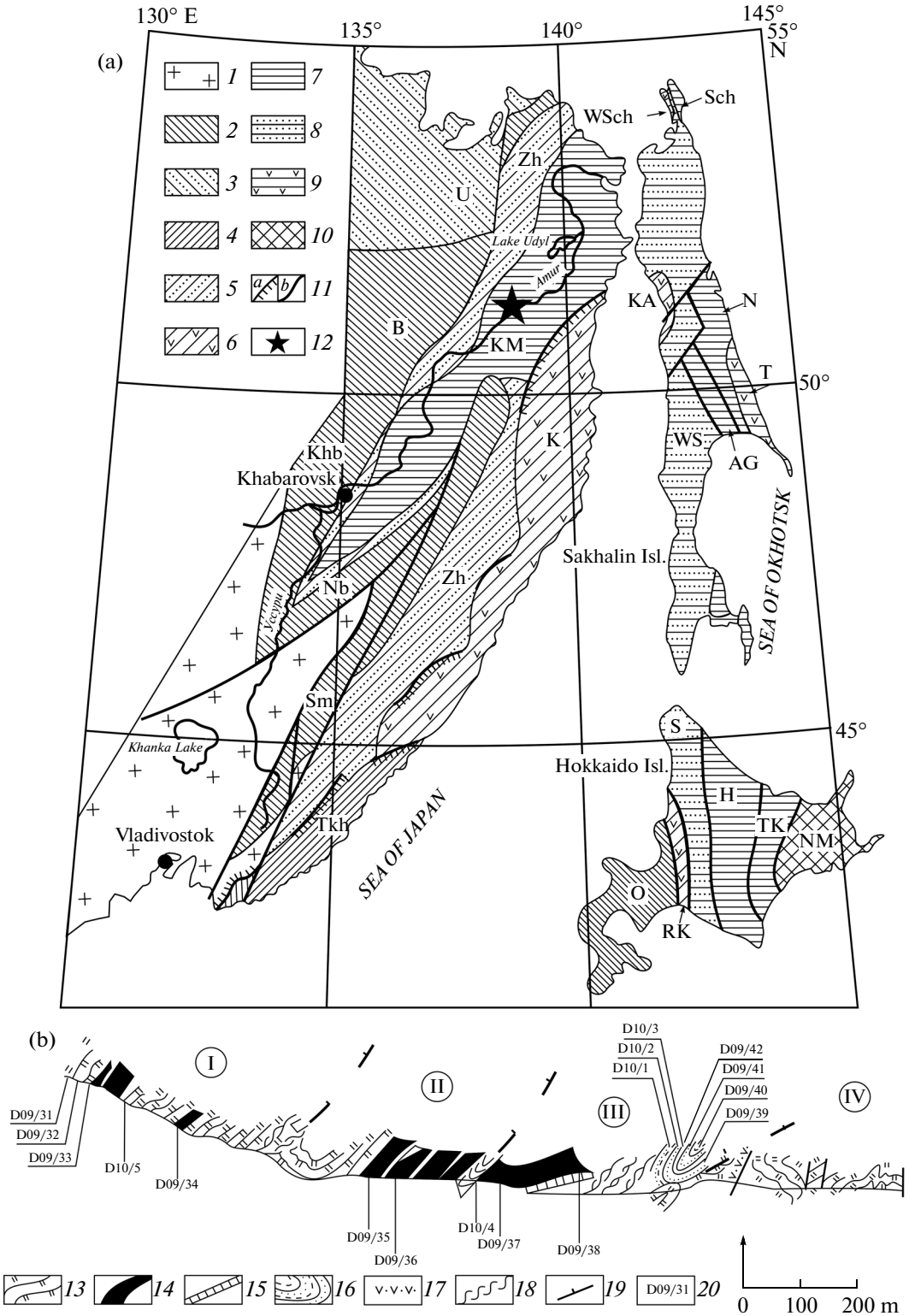


Fig. 1. The tectonostratigraphic map of terrains of the Sikhote-Alin' Region and adjacent regions ([10, 11], modified). (*1*) Pre-Mesozoic; (*2, 3*) Jurassic: (*2*) accretionary prism (B—Badzhal, NB—Nadanhada-Bikin, O—Oshima, Sm—Samarka, Khb—Khabarovsk), (*3*) turbidite (U—Ulban); (*4–6*) Early Cretaceous: (*4*) accretionary prism (Tkh—Taukhe), (*5*) turbidite (Zh—Zhuravlevka-Amur), (*6*) island arc (K—Kema, KA—Kamyshovyi, RK—Rebun-Kabato, WSch—West Schmidt); (*7–9*) Early Cretaceous–Late Cretaceous: (*7*) accretionary prism (AG—Aniva-Gomon, KM—Kiselevka-Manoma, N—Nabil, TK—Tokoro, H—Hidaka, Sch—Schmidt), (*8*) turbidite (WS—West Sakhalin, S—Sorachi-Yeso), (*9*) island arc (T—Terpeniya); (*10*) Late Cretaceous–Paleogene (NM—Nemuro); (*11*) faults: *a*—thrusts, *b*—oblique-slip faults; (*12*) area of detailed paleomagnetic studies ([2, 4], modified); (*13*) cherts; (*14*) basalts; (*15*) limestones; (*16*) turbidites with chert fragments; (*17*) tuffs; (*18*) schistosity zones; (*19*) tectonic boundaries; (*20*) sample sites and numbers of paleomagnetic samples; roman numerals—numbers of tectonic plates.

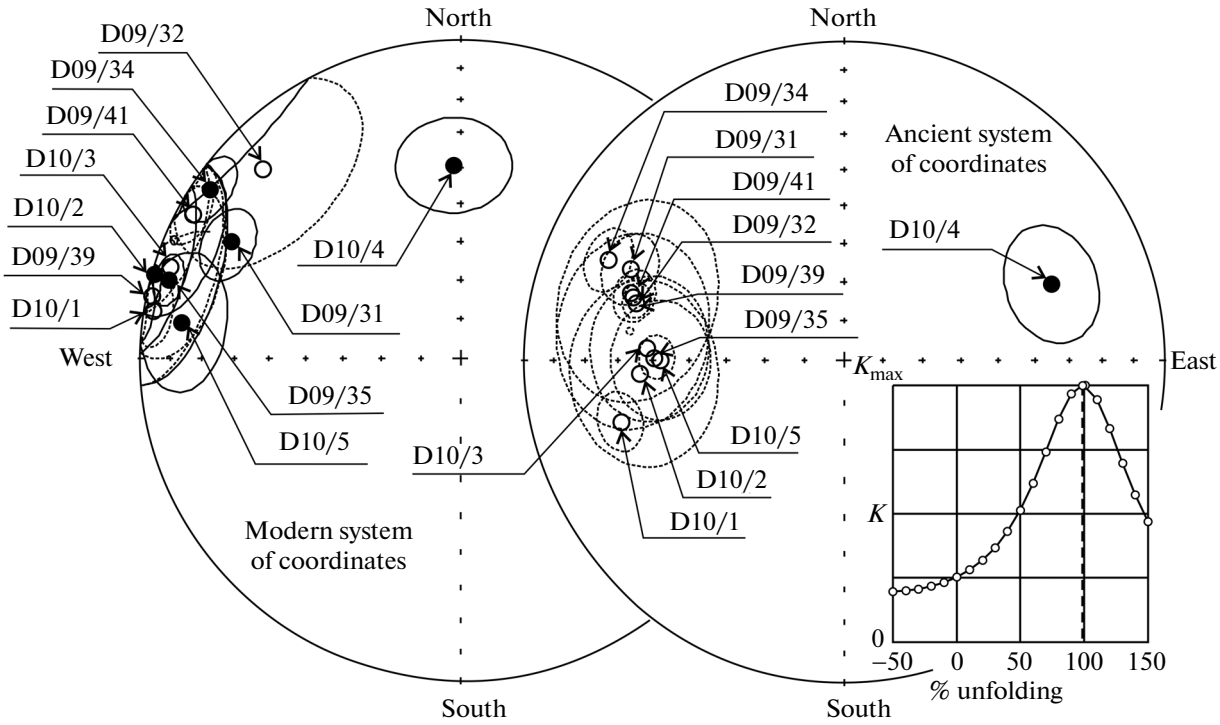


Fig. 2. Stereograms of distribution of high-temperature NRM component directions of rocks of the Kiselevka complex (ancient and modern systems of coordinates). Black and open symbols represent projections on the lower and upper semi-spheres, correspondingly. The plot in the lower right-hand corner represents step unfolding.

ture and high-temperature) of natural remanent magnetization (NRM) were revealed.

The low-temperature component was revealed by thermal cleaning of 86 samples in the temperature range between 20 and 250–350°C. In the geographical system of coordinates Dec = 350.8°, Inc = 66.4° (K = 9.3, α_{95} = 5.3°).[1]

In the stratigraphic system of coordinates, the K parameter is rather lower (3.2). The direction of the low-temperature NRM component in the geographical system of coordinates is similar to the present-day geomagnetic field direction in the area of study (Dec = 347.8°, Inc = 66.1°).

The high-temperature component was revealed by thermal cleaning of 73 samples, collected at 11 sampling sites, in the temperature range between 400 and 560–620°C. As a result of thermomagnetic cleaning, the NRM component points for the most of the samples on Zijderveld diagrams trends towards the origin of the Zijderveld diagram; i.e., it can be called the characteristic remanent magnetization. Figurative points are in third and fourth quadrants of the stereogram; only samples from the D10/4 site are in the first quadrant (Fig. 2). We suggest that magnetization of these samples occurred in a reverse magnetic field, as opposed to magnetization of samples from the other 10 samples.

Therefore, when calculating the mean paleomagnetic directions for the high-T component, the directions for samples collected at the D10/4 site were converted into another polarity.

In the geographical and stratigraphic systems of coordinates, the mean directions in the high-T NRM component in all 73 samples are Dec_g = 285.9°, Inc_g = −0.4° (K_g = 5.8, α_{95} = 6.8°) and Dec_s = 265.8°, Inc_s = −34.3° (K_s = 11.3, α_{95} = 4.9°), respectively.

In the stratigraphic system of coordinates concentration the K value is twice as high compared to geographical system of coordinates. The average direction for 11 points in the geographical and stratigraphic systems is Dec_g = 286.4°, Inc_g = −2.1° (K_g = 6.3, α_{95} = 19.8°) и Dec_s = 263.6°, Inc_s = −33.8° (K_s = 33.3, α_{95} = 8.0°), respectively.

In the second case, the K value in the stratigraphic system of coordinates is 5 times higher than in the geographical system of coordinates. Here, in the steps of the unfolding procedure [13], the K parameter is maximum at 100% unfolding (Fig. 2), which is evidence of the prefolding age of the high-T component. Taking into account directly magnetized (one sampling site–7 samples) and reversely magnetized (10 sampling sites–66 samples) rock samples, one can confidently

[1] Dec, Inc–paleomagnetic declination and inclination, correspondingly. K–Fisher precision parameter; α_{95}—a radius of the cone about the mean direction at the probability of ρ = 0.95.

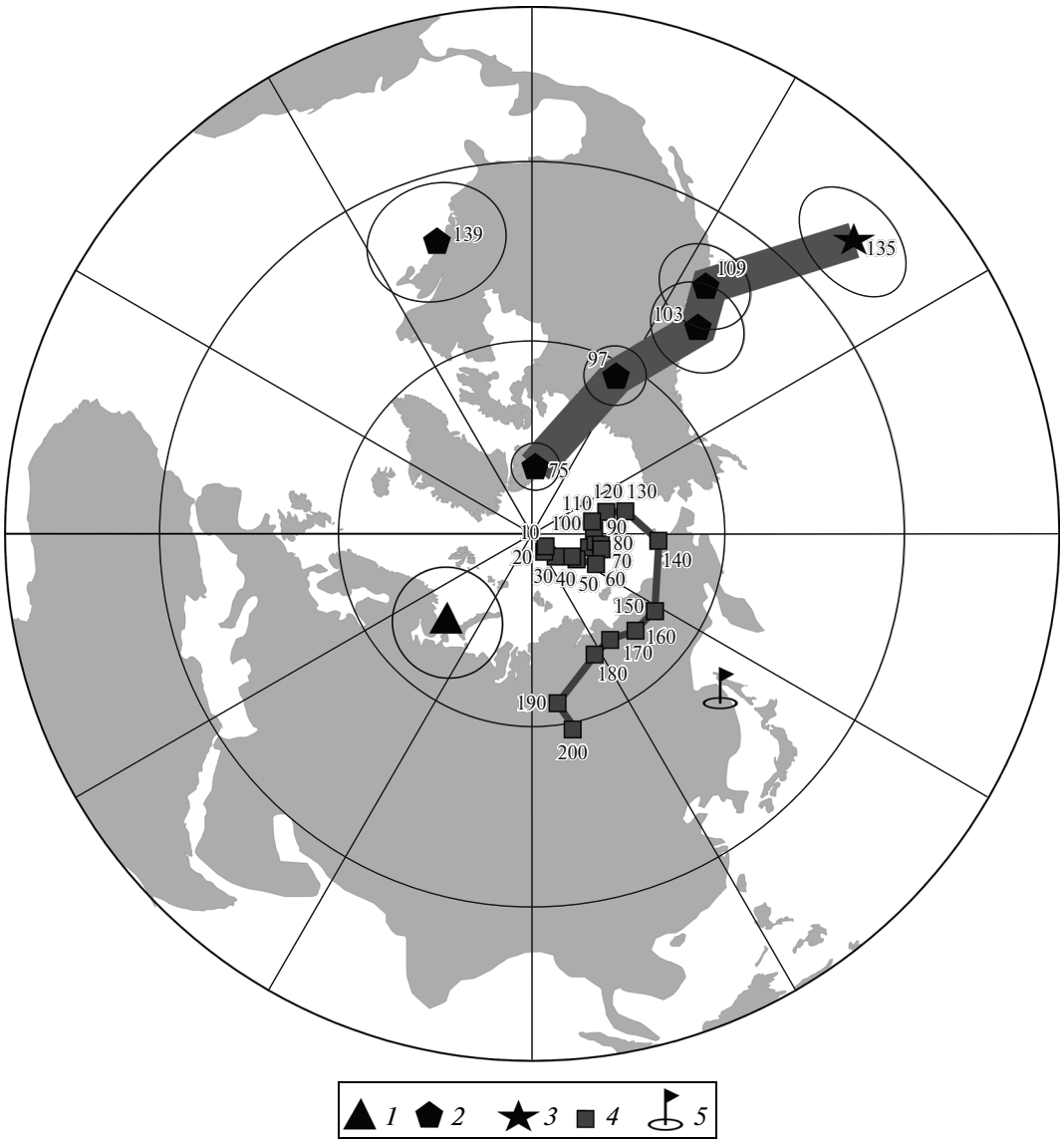


Fig. 3. Positions of paleomagnetic poles calculated for Mesozoic rocks of the Sikhote-Alin–North Sakhalin orogenic belt. (*1*) The pole of the Late Cretaceous metachronic component [1]; (*2*) paleomagnetic poles of Cretaceous rocks of the West Sakhalin trough [12]; (*3*) the pole of Lower Cretaceous rocks of the Kiselevka terrain (this work); (*4*) Mesozoic–Cenozoic fragments of the apparent polar wander path (APWP) for stable Eurasia [15]; (*5*) the area of study. Circles around the poles are confidence circles at the probability of $\rho = 0.05$. Numbers near the poles are age data, Ma.

suggest that the age of the high-temperature component is close to the time of formation of the rocks studied (nearly 135 Ma).

The coordinates of the paleomagnetic pole determined from Cretaceous rocks of the Kiselevka complex are as follows: Plat = 18.6°, Plong = 222.4°, dp = 5.2°, and dm = 9.1°. This pole is located in the tropical latitudes of the Northern Hemisphere (the northern part of the Pacific) (Fig. 3). In addition, Fig. 3 shows the position of the paleomagnetic pole, corresponding to regional remagnetization [1], the positions of paleomagnetic poles for Cretaceous rocks of the West Sakhalin trough [12], and the Mesozoic–Cenozoic fragment of the trajectory of the apparent polar wander path (APWP) of the stable Eurasian continent (Europe) [15]. Moreover, one can see (Fig. 3) that the paleomagnetic pole determined for Lower Cretaceous rocks of the Kiselevka complex and paleomagnetic poles for Cretaceous rocks of Sakhalin Island are located far from the direction of probable remagnetization [1] and APWP of stable Eurasia [15], and poles from the Early Cretaceous to the Late Cretaceous are located in the submeridional zone from tropical to high latitudes of the Northern Hemisphere. In our

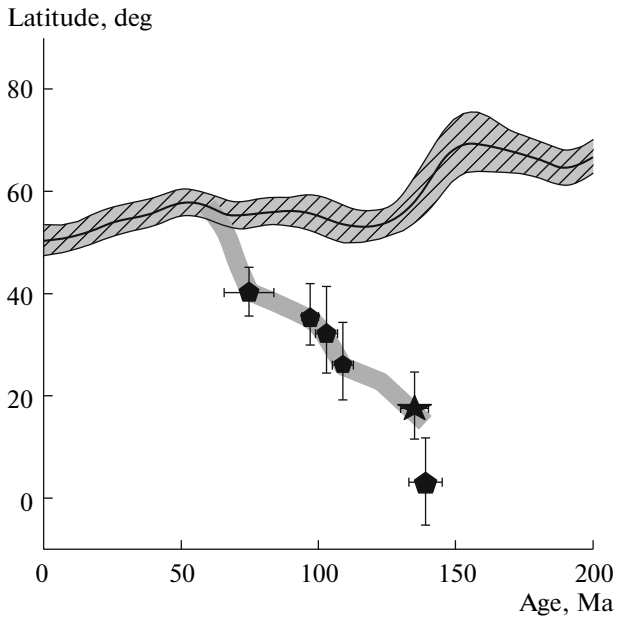


Fig. 4. Comparison of paleolatitudes (crosshatched band) calculated from APWP for the stable Eurasian continent [15] with revealed paleolatitudes (unhatched gray band) in Cretaceous rocks of the West Sakhalin trough [12] and the Kiselevka complex (this work).

opinion, such a regular position of the poles is evidence of the general evolution of large-amplitude horizontal displacements relative to the stable position of Eurasia in the Cretaceous.

This general trend is seen distinctly in the dependence plot between paleolatitudes calculated from APWP for the stable Eurasian continent in the Cretaceous and the revealed paleolatitudes in Cretaceous rocks of the West Sakhalin trough [15] and the Kiselevka complex in the age interval of 0–200 Ma (Fig. 4). As seen in Fig. 4, the Cretaceous complexes studied in Primorie and Sakhalin displaced gradually from equatorial-tropical to modern latitudes of the Northern Hemisphere towards the stable Eurasian continent in the age range of 80–60 Ma.

Based on results of the analysis of paleomagnetic data for Cretaceous rocks of complexes of Sakhalin [12] and Primorie, geological–geochemical data on their genesis [2], and the available paleotectonic [3, 10] and global kinematic [14] reconstructions, the next model of evolution of the region in the Cretaceous is proposed.

The volcanogenic–sedimentary rocks of the Kiselevka complex formed nearly 135 Ma ago in the latitudinal zone of 18° ± 5° of the Northern Hemisphere within the Isanagi oceanic plate, which was displaced in the northwestern direction with a rate of approximately 20 cm/year and submerged beneath the eastern margin of Eurasia. The conclusion, made on the basis of new paleomagnetic data, is in agreement with available lithological and geochemical data. According to the latter, radiolarites of the Kiselevka block were deposited in deep-water settings; basalts correspond to enriched intraplate basalts from a mantle plume source, like a hot spot of the Hawaiian Islands [2, 4, 11]. The modern coordinates of the Hawaiian Islands are 19°–20°N.

In the Late Early Cretaceous (105 ± 5 Ma), the direction of movement of the Isanagi oceanic plate changed into nearly submeridional; the movement rate decreased to 10–15 cm/year.

By that time, the Kiselevka complex on the Isanagi plate was displaced in northwesterly direction up to 40° latitude, reached the margin of the stable Eurasian continent near the coast of the Korean Peninsula, and joined the Kiselevka-Manoma accretionary prism. In turn, the latter was a part of the syn-shear near-continental zone (transform boundary?) and was displaced in northerly–northeasterly direction along the continental margin.

At the end of the Late Cretaceous (70 ± 5 Ma), the direction of movement of the Isanagi plate changed to northwesterly; the rate of movement increased sharply up to 25 cm/year. By that time, the Kiselevka block, as a part of the Kiselevka-Manoma accretionary prism, was displaced along the eastern margin of Eurasia, as a fact, as far as its modern position (Lower Amur), where it joined the Eurasian continental plate.

The paleomagnetic data obtained are evidence of large-scale displacement of individual blocks, comprising the Sikhote-Alin'–North Sakhalin orogenic belt. These data are in agreement with the terrain model proposed for this orogenic belt. The value of the northern component of displacement of the Kiselevka terrain is nearly 33° (more than 3500 km), that is, from +18° ± 5 ° in the Early Cretaceous to +51.4° at present.

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